Board Sheet — Lesson 8: Linear Regression I

MA153X Mathematical Modeling

Name: ______________________________ **Date:** ________________

Key Vocabulary (from Reading 2.2)

Simple linear regression: ____

Independent variable (x): ____

Dependent variable (y): ____

Error term (ε): ____

Sum of Squared Errors (SSE): ____

The Model

The **population model** is:

$$y = \_\_\_\_\_\_\_\_ + \_\_\_\_\_\_\_\_ \cdot x + \_\_\_\_\_\_\_\_$$

The **estimated (fitted) model** is:

$$\hat{y} = \_\_\_\_\_\_\_\_ + \_\_\_\_\_\_\_\_ \cdot x$$

Where:

- $\hat{y}$ represents: ______________________________
- $\hat{\beta}_0$ represents: ______________________________
- $\hat{\beta}_1$ represents: ______________________________

Four Assumptions of SLR (from Reading 2.2)

Fill in each assumption:

1. **Linearity:** ______________________________
2. **Independence:** ______________________________
3. **Constant variance:** ______________________________
4. **Normality:** ______________________________

How do we justify the linearity assumption? ______________________________

Training Data (ACFT — 20th Engineer Brigade)

Can we predict a soldier's weight from their height?

Soldier	Height (in)	Weight (lbs)
S002	74	239
S003	63	147
S004	70	169
S005	68	159
S006	72	211
S007	73	239
S008	71	202
S009	64	136
S010	67	194
S011	74	208

(Full training set of 37 soldiers is in the Excel workbook)

Excel Practice

1. What is the independent variable (x)? ____________________

2. What is the dependent variable (y)? ____________________

3. Create a scatter plot in Excel. Does the relationship appear linear? ________

4. Method 1 — Trendline: Add a linear trendline. Record the equation:

$$\hat{y} = \underline{\hspace{3cm}} + \underline{\hspace{3cm}} \cdot x$$

5. Method 2 — Solver: Set up columns for x, y, $\hat{y}$, error, squared error. Use Solver to minimize SSE.

- $\hat{\beta}_0$ from Solver: ____________________
- $\hat{\beta}_1$ from Solver: ____________________

- SSE at solution: ______________________

6. **Verify** using Excel formulas:

- `=SLOPE(...)` → ______________________
- `=INTERCEPT(...)` → ______________________

7. Do all three methods agree? ____________

Making a Prediction

A soldier is 70 inches tall. Predict their weight:

Takeaways

What does simple linear regression do?

Why must we justify the choice of a linear model before fitting it?
